

## SOFTWARE REQUIREMENTS SPECIFICATION

# Smart Home Maintenance Services

*An online marketplace for home repair and maintenance services, with call-verified payment release*

| Field                     | Detail                                                            |
|---------------------------|-------------------------------------------------------------------|
| Project title             | Smart Home Maintenance Services                                   |
| Project domain / category | Web Programming                                                   |
| Document                  | Software Requirements Specification (SRS)                         |
| Version                   | 2.0                                                               |
| Date                      | 4 September 2026                                                  |
| Basis document            | FR.pdf — Functional Requirements, Smart Home Maintenance Services |

*This specification supersedes the functional requirements in FR.pdf. Every requirement stated in that document is carried forward here and is traceable through Appendix A. Requirements added by this revision fill structural gaps identified during analysis or reflect current practice in the on-demand home services category.*

## Revision History

| Version | Date       | Author        | Description                                                                                                                                                                                                                          |
|---------|------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | —          | Project brief | Original functional requirements (FR.pdf): four modules — Service Provider, Customer, Admin, Payment.                                                                                                                                |
| 2.0     | 4 Sep 2026 | Project team  | Full SRS. Gap analysis of v1.0; catalogue, availability, booking lifecycle, evidence capture, escrow and verification added. Payment release made conditional on a telephone verification call. Conduct and penalty framework added. |

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# 1. Introduction

## 1.1 Purpose

This document specifies the complete software requirements for Smart Home Maintenance Services, a web-based marketplace that connects households with verified tradesmen — plumbers, electricians, sanitary workers, appliance technicians and related trades — and manages the engagement end to end: discovery, booking, execution, verification of the completed work, payment, and dispute handling.

The document is written to be directly buildable. Every requirement carries a stable identifier, a priority, and a source, so it can be traced into the design, the implementation and the test plan.

## 1.2 Document Conventions

- Requirement identifiers follow the pattern FR-XX-nn for functional requirements and NFR-XX-nn for non-functional requirements, where XX identifies the module.
- Priority is High (required for a working system), Medium (required for a complete system), or Low (desirable).
- Source is either FR.pdf (the requirement appears in the original brief) or New (added by this revision).
- System states, database tables and field names are written in a monospaced typeface.
- The words shall and must denote mandatory behaviour; may denotes optional behaviour.

## 1.3 Intended Audience

| Reader                         | Suggested reading order                                                               |
|--------------------------------|---------------------------------------------------------------------------------------|
| Project supervisor / evaluator | Sections 1, 2, 4 and 7, then Appendix A for traceability against the original brief.  |
| Developer                      | Sections 3, 5, 6, 10 and 11, then Section 13 for build sequence.                      |
| Database designer              | Sections 5 and 10, then the integrity rules in 10.3.                                  |
| Tester                         | Section 6 (requirement IDs), Section 5 (state transitions) and Section 8 (use cases). |
| Operations staff               | Sections 4, 7 and 9.1.                                                                |

## 1.4 Project Scope

The system is a three-sided web application. Customers search for and book tradesmen; tradesmen (referred to throughout as service providers) publish their skills and availability and carry out the work; platform staff approve providers, verify completed work by telephone, release payment, and resolve disputes.

The defining characteristic of this system is that payment is not released on the tradesman's word alone. On completion, the job enters a verification queue; the platform telephones the customer, records structured feedback about the workman, and only then releases the money. This single rule determines the payment architecture (funds must be held), the staffing model (a verification role exists), and the integrity of the rating system (a rating can only originate from a verified call).

In scope: web application for all three user classes, service catalogue, availability and slot booking, job execution evidence, verification, escrow and payouts, ratings, complaints and disputes, notifications, maintenance plans, administrative reporting, and a conduct and penalty framework.

Out of scope for this release: native mobile applications, live GPS tracking, AI-based matching, video quotation, dynamic surge pricing, insurance products, and multi-city operations. Section 12 records the reasoning.

## 1.5 Definitions, Acronyms and Abbreviations

| Term                       | Meaning                                                                                                                          |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Service provider / workman | A registered tradesman who performs the work. Used interchangeably in this document.                                             |
| Verification call          | The outbound telephone call placed by platform staff to the customer after work completion, before payment is released.          |
| Verification agent         | The staff member who places verification calls and records the structured feedback.                                              |
| Escrow                     | Funds captured from the customer and held by the platform until a verification outcome permits release.                          |
| Evidence floor             | The set of records automatically captured for every job: start OTP, geofence times, photographs, checklist and itemised invoice. |
| Start OTP                  | A one-time code shown on the customer's phone that the provider must enter to begin work; it evidences arrival.                  |
| Rework                     | Work returned to the provider for correction at no additional charge, with funds still held.                                     |
| Auto-release               | Automatic release of held funds when the customer cannot be reached within the defined window.                                   |
| Demerit points             | Units recorded against a provider for verified breaches; accumulation triggers graduated consequences.                           |
| AMC                        | Annual maintenance contract — a recurring maintenance plan.                                                                      |
| SLA                        | Service level agreement — a time limit the platform commits to.                                                                  |
| OTP                        | One-time password.                                                                                                               |
| CNIC                       | Computerised National Identity Card.                                                                                             |

## 1.6 References

- FR.pdf — Functional Requirements, Smart Home Maintenance Services (Web Programming). The basis document for this specification.
- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications. Structure of this document.
- Acquaint Softtech, "Building a Home Services Marketplace App" — reference for standard module decomposition in this category.
- Megabant, "Top 7 Home Services Marketplace Trends to Watch in 2026" — reference for current practice.



## 2. Overall Description

### 2.1 Product Perspective

The system is a new, self-contained web application with no predecessor. It is the intermediary in every transaction rather than a directory: it holds the money, records the evidence, verifies the outcome and carries the reputation data. This is what distinguishes it from a classified listing site, and it is the reason the payment, evidence and verification modules cannot be treated as optional extras.

External systems it depends upon: a payment gateway for card and mobile-wallet capture, an SMS gateway for OTP and notifications, an email service, and a mapping service for address geocoding and distance calculation.

### 2.2 Product Functions

The system is decomposed into fifteen functional modules.

| Module | Function                                                                                           |
|--------|----------------------------------------------------------------------------------------------------|
| M1     | Service catalogue and pricing — the definition of what can be booked and how it is priced.         |
| M2     | Customer — registration, profile, addresses, booking history.                                      |
| M3     | Service provider — registration, documents, expertise, availability, earnings.                     |
| M4     | Search and matching — service-first discovery, filtering and ranking.                              |
| M5     | Booking and scheduling — slot selection, acceptance, rescheduling, cancellation, state history.    |
| M6     | Work execution and evidence — arrival OTP, photographs, checklists, revised quotes, invoicing.     |
| M7     | Verification and feedback — the call queue, agent console, questionnaire and outcomes.             |
| M8     | Payments, escrow and payouts — capture, hold, release, commission, refunds, provider payouts.      |
| M9     | Ratings and reputation — scores and remarks derived exclusively from verified calls.               |
| M10    | Complaints and disputes — lifecycle, evidence, right of reply, resolution outcomes.                |
| M11    | Notifications — email, SMS and in-app messaging across every state change.                         |
| M12    | Administration and operations — approvals, blocking, roles, settings, audit log, operations board. |
| M13    | Maintenance plans — recurring maintenance subscriptions and entitlements.                          |
| M14    | Reporting — monthly, revenue, provider performance and verification reports.                       |
| M15    | Conduct and penalties — the demerit schedule, thresholds, appeals and incentives.                  |

### 2.3 User Classes and Characteristics

| User class | Characteristics                                                                                                                                      | Technical skill |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Customer   | Householder needing repair or maintenance. Uses the system occasionally and usually under time pressure. Accesses predominantly from a mobile phone. | Low to moderate |

| User class         | Characteristics                                                                                                                                                          | Technical skill |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Service provider   | Tradesman, often self-employed. Uses the system daily. Almost always on a low-cost Android phone, frequently on a slow connection. May have limited literacy in English. | Low             |
| Verification agent | Platform staff. Uses one screen intensively for an entire shift; efficiency of that screen determines operating cost.                                                    | Moderate        |
| Finance officer    | Platform staff. Releases escrow, processes refunds, runs payout batches and reconciles cash jobs.                                                                        | Moderate        |
| Administrator      | Platform owner or manager. Approves providers, resolves disputes, configures the system and reads reports.                                                               | High            |

## 2.4 Operating Environment

- Client: any modern web browser on desktop, tablet or mobile. The interface is mobile-first and must remain fully usable at 360 px width.
- Server: a standard web application server with a relational database. The application is not tied to a specific stack; any server-side language with a relational database and a job queue satisfies the requirements.
- Network: the provider-facing interface must remain usable on 3G connections; image upload must tolerate interruption and resume or retry.
- Language: English and Urdu, with right-to-left support for Urdu text.

## 2.5 Design and Implementation Constraints

- All monetary movements must occur inside database transactions and be recorded in a double-entry ledger. Balances are derived from the ledger and are never stored as standalone editable values.
- Payment gateway callbacks must be idempotent — a repeated callback must never release or refund twice.
- Verification records are write-once. Corrections are stored as linked amendments, never as edits.
- The audit log is append-only.
- Account removal is a soft deactivation. No user record that has participated in a booking may be physically deleted, because the financial and reputation history depends on it.
- All operational timings, fees, thresholds and windows are held in configuration, not in code.

## 2.6 Assumptions and Dependencies

- Every customer registers with a mobile number that can receive calls and SMS; this is the number the verification call is placed to, and the system cannot function without it being genuine.
- Providers own a smartphone with a camera capable of capturing job photographs.
- Cash remains a common payment method and must be supported as a first-class path, not an exception.
- A payment gateway supporting card and mobile-wallet transactions is available and permits held funds or delayed capture.
- Verification agents are available during defined calling hours; the system does not place calls outside 08:00–22:00.



### 3. Service Catalogue

The original brief names sanitary work, plumbing and electrical repair and leaves the remainder open. This section defines the catalogue concretely, because nothing in the system can be searched, priced, booked or reported on until services exist as records.

#### 3.1 Categories and Services

| Category                         | Services                                                                                                                                                                                                  |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Plumbing                         | Tap and mixer replacement; pipe leak repair; water tank cleaning; motor and pump repair or installation; drain unblocking; water filter installation; geyser installation and servicing.                  |
| Sanitary and bathroom            | Commode installation or replacement; flush tank repair; washbasin fitting; shower and bath fitting; bathroom tile and seal repair; sewerage line blockage.                                                |
| Electrical                       | Switch, socket and distribution board repair; wiring fault tracing; fan installation and repair; light and chandelier fitting; UPS and inverter installation; voltage stabiliser fitting; house rewiring. |
| Appliance repair                 | AC servicing; AC gas refilling; AC installation and dismantling; refrigerator repair; washing machine repair; microwave repair; water dispenser repair.                                                   |
| Carpentry and furniture          | Door and lock repair; window and hinge repair; cupboard and wardrobe repair; furniture assembly; curtain rod and shelf fitting; kitchen cabinet repair.                                                   |
| Paint, masonry and waterproofing | Interior and exterior painting; wall crack and plaster repair; roof waterproofing; tile fixing; false ceiling repair; damp treatment.                                                                     |

Two further categories are supported by the catalogue structure but are held back from the first release, on the reasoning that deep coverage in a few categories serves customers better than thin coverage in many: cleaning and pest control (deep cleaning, water tank disinfection, fumigation), and security and smart home (CCTV, video door phone, WiFi installation, solar panel cleaning).

#### 3.2 Pricing Models

Each catalogue entry is priced under one of three models. The third is the one the rest of the system must accommodate, because it is the origin of on-site quote revisions and therefore of most payment disputes.

| Model            | Applies to                                                                                                                                                                            | Examples                                                                      |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Flat rate        | Standardised jobs with a predictable scope.                                                                                                                                           | AC servicing; fan installation; water tank cleaning.                          |
| per Day          | Open-ended labour where the duration cannot be known in advance.                                                                                                                      | General plumbing call-out; carpentry; painting labour.                        |
| Inspection first | Faults that cannot be diagnosed without a visit. A visit fee is charged, the provider diagnoses, and a revised quote is raised for the customer's in-app approval before work begins. | Refrigerator repair; wiring fault tracing; damp treatment; sewerage blockage. |

#### 3.3 Service Attributes

| Attribute              | Meaning and typical use                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Emergency eligible     | May be booked as same-day or out-of-hours with a configured surcharge. Applies to a small subset only: burst pipe, total power failure, sewerage overflow, geyser leak. |
| Plan eligible          | May be included in a recurring maintenance plan. Applies to scheduled and seasonal work: AC servicing, water tank cleaning, plumbing inspection.                        |
| Warranty eligible      | Carries a workmanship warranty of a configured length. Applies where failure would appear within days: leak repairs, appliance repairs, electrical fault work.          |
| Verification tier bias | Marks high-value or hard-to-judge services that always route to a human verification call — for example rewiring, waterproofing and AC gas refilling.                   |

## 4. Work Verification and Payment Release

This section specifies the mechanism that distinguishes this system. It is placed before the functional requirements because several modules exist only to serve it.

### 4.1 Rationale

The original brief states that responsibility for verifying completed work rests with the customer, but provides no mechanism. In practice a rating entered into an application while the tradesman waits in the room is not an independent rating, and a payment handed over at that moment is not an informed payment.

Moving feedback to a telephone call placed by the platform after the tradesman has left, with the money still held, removes that pressure and produces a verified record of every completed job. It also makes fabricated reviews structurally impossible: a rating can exist only where a completed verification call exists.

### 4.2 The Evidence Floor

The following are captured automatically for every job, at no marginal cost, before any verification decision is taken. Unlike the call, they work when the customer is not present, and they are the evidence any later dispute is decided on.

| Evidence                                                    | What it establishes                                                                                                  |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Start OTP entered by the provider from the customer's phone | That the provider physically arrived and that the customer was present at the start.                                 |
| Geofenced check-in and check-out                            | Location of the work and total time on site, comparable against the service's expected duration.                     |
| Before and after photographs, timestamped                   | The condition of the work before and after; the primary evidence in a dispute.                                       |
| Completed service checklist                                 | That each defined step of that specific service was performed, rather than a single unqualified claim of completion. |
| Itemised invoice with approved extras                       | What was charged and whether each addition was approved by the customer in advance.                                  |

### 4.3 Verification Tiers

Placing a staffed call on every job does not scale: an agent handles roughly fifty to sixty calls in a shift, so a thousand jobs a day would require seventeen agents. Deterrence, however, derives from probability rather than coverage — a provider who knows that a call may come, and cannot predict which job it will come for, behaves as though every job will be called. The system therefore routes each completed job to one of two tiers by rule.

| Tier   | Method                                     | Applied when                                                                                                                                                                                                                                                                                                 |
|--------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tier A | Verification call placed by a human agent. | The provider's first five jobs; job value above the configured threshold; a service marked high-risk; an evidence anomaly (time on site far below expectation, missing photographs, geofence mismatch); the provider currently carries demerit points; a prior dispute exists between this customer and this |

| Tier   | Method                                                                          | Applied when                                                                                         |
|--------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|        |                                                                                 | provider; extras were added on site; or the job falls in the configured random sample (default 10%). |
| Tier B | Automated confirmation by WhatsApp or SMS one-tap link, or IVR keypad response. | All other jobs.                                                                                      |

Both tiers write to the same verification record and produce the same outcome set, so all downstream logic is identical. A Tier B job that receives no response within 24 hours is escalated to Tier A. The tier applied is stored on the booking so that coverage can be reported on.

**Implementation note.** The routing rules should be implemented from the outset but shipped with thresholds set so that every job routes to Tier A. This demonstrates the full call workflow while leaving a documented, configuration-level answer to the question of how the system behaves at volume.

#### 4.4 The Verification Sequence

| State                        | What happens                                                                                                                                                                                                                                              |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WORK_COMPLETED               | The provider marks the job complete, supplying after-photographs, parts used and an itemised final amount. Completion is blocked unless the start OTP was previously entered.                                                                             |
| AWAITING_VERIFICATION        | The booking enters the verification queue and all funds freeze. Prepaid funds remain in escrow; for a cash job the customer is instructed by SMS to withhold payment until the call. SLA: contact within 30 minutes, within calling hours of 08:00–22:00. |
| CALL_IN_PROGRESS             | The agent console presents the booking, provider history, photographs and itemised amount on a single screen. The agent works through a fixed questionnaire; the call is recorded following a spoken consent line.                                        |
| VERIFIED / REWORK / DISPUTED | The agent submits one of the four outcomes in 4.6. The record becomes immutable at this point.                                                                                                                                                            |
| PAYMENT_RELEASED             | Escrow moves to the provider's wallet less commission, or the cash job is marked settled and commission is debited. The rating publishes at this moment and not before.                                                                                   |

#### 4.5 Feedback Captured on the Call

Feedback is captured on a fixed questionnaire. Free conversation is not recorded as feedback, because it cannot be aggregated or compared.

- Was the work completed — fully, partially, or not at all?
- Quality of work, punctuality, conduct, and cleanliness of the site on leaving — each rated 1 to 5. The published star rating is the average of these four.
- Was any amount demanded beyond the quoted or approved price? This single question is the platform's principal defence against off-book charging, and no automated mechanism can substitute for it.
- Did the workman arrive in uniform and carry his own tools?

- Verbal consent to release payment — yes or no.
- A free-text remark, transcribed by the agent, published as the customer's remark on the provider's profile.
- Agent identifier, call duration, recording reference and timestamp, recorded automatically.

#### 4.6 Verification Outcomes

| Outcome               | Effect                                                                                                                                                                                                                    |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verified — satisfied  | Consent given. Payment releases immediately; the rating and remark publish; the provider's completed-job count increments.                                                                                                |
| Verified — with issue | The work is accepted but a concern was voiced. Payment releases, a complaint ticket opens automatically, and the provider is flagged. Three flags within 90 days trigger administrative review.                           |
| Rework required       | Funds remain held. The provider is given a 48-hour rework window at no additional charge; on re-completion the booking re-enters the queue. Two failed reworks escalate to a dispute.                                     |
| Disputed              | The work is denied, or an unapproved charge was demanded. Funds freeze and the case enters the administrator's dispute queue, resolving as full release, partial release, full refund, or refund with a provider penalty. |

#### 4.7 Unreachable Customer

An unreachable customer must not indefinitely withhold a tradesman's earnings; this is the failure mode that makes naive escrow designs unusable in practice.

1. Three call attempts are made across two different time bands within 24 hours, each logged individually.
2. If all three fail, an SMS and WhatsApp link is sent offering one-tap OTP confirmation.
3. If there is still no response 72 hours after completion, the job auto-releases under a published no-response policy: payment is released and the completion is recorded, but no rating is published — an unrated job neither inflates nor damages the provider's score.
4. The customer retains a seven-day window after auto-release in which to raise a complaint.

#### 4.8 Cash Jobs

The original brief permits payment in cash on the spot, and cash is expected to remain the majority payment method. The order of events is therefore reversed rather than the path being excluded: the verification call takes place before money changes hands. The agent confirms the work and states the exact payable amount, the customer pays in cash, the provider confirms receipt in the application, and the platform commission is debited from the provider's wallet or netted against his next payout. A provider whose commission debt exceeds the configured ceiling receives no further job offers until it is cleared.

#### 4.9 Alternative Mechanisms Considered

The following were evaluated as substitutes for the verification call and are recorded here so the design decision is auditable.

| Mechanism                                           | Assessment                                                                                                                                                                                     |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Completion OTP only                                 | Free and instant, but obtained with the workman present, and impossible when the customer is absent. Retained as part of the evidence floor, rejected as the verification mechanism.           |
| In-app confirmation button with auto-approval timer | Removes on-site pressure but attracts low response rates; in practice most releases occur on the timer. Retained as the Tier B fallback.                                                       |
| Automated IVR call with keypad rating               | Roughly a tenth of the cost of a staffed call and scales without limit, but cannot ask the follow-up question about additional charges or capture a usable remark. Adopted as a Tier B option. |
| Supervisor spot-audit                               | Effective for quality assessment and adopted as an administrative control, but too slow and costly to gate every payment.                                                                      |
| Dispute window with immediate release               | Zero friction and zero cost, but recovers money only by clawback, which is impractical against self-employed tradesmen. Adopted only as the auto-release fallback in 4.7.                      |
| Retention against warranty                          | Holding back a percentage until the warranty expires aligns incentives with durable work. Recorded as a future enhancement rather than a first-release requirement.                            |
| IoT and sensor telemetry                            | Verifies the outcome rather than the visit — for example a temperature differential after AC servicing. Narrow in application; recorded in Section 12 as a future enhancement.                 |

## 5. Booking Lifecycle

Every booking is a state machine. No process may alter a booking except through a named transition, and every transition writes a row to the booking status history recording actor, timestamp and reason.

### 5.1 Principal Path

REQUESTED → ACCEPTED → SCHEDULED → EN\_ROUTE → IN\_PROGRESS → WORK\_COMPLETED →  
AWAITING\_VERIFICATION → VERIFIED → PAYMENT\_RELEASED → CLOSED

### 5.2 States

| State                 | Meaning                                                                      |
|-----------------------|------------------------------------------------------------------------------|
| REQUESTED             | The customer has submitted a booking; the provider has not yet responded.    |
| ACCEPTED              | The provider has accepted the job.                                           |
| SCHEDULED             | The slot is confirmed and locked on the provider's calendar.                 |
| EN_ROUTE              | The provider has set out; the customer has been notified.                    |
| IN_PROGRESS           | The start OTP has been entered and work has begun.                           |
| WORK_COMPLETED        | The provider has submitted completion evidence and the final amount.         |
| AWAITING_VERIFICATION | The job is in the verification queue; all funds frozen.                      |
| VERIFIED              | A verification outcome permitting release has been recorded.                 |
| PAYMENT_RELEASED      | Funds have moved to the provider's wallet, or the cash job has been settled. |
| CLOSED                | The job is complete and the warranty period has elapsed with no claim.       |

### 5.3 Branch Transitions

| From                  | To                 | Rule                                                                                                                             |
|-----------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| REQUESTED             | DECLINED / EXPIRED | The provider declines, or does not respond within 15 minutes. The request is offered automatically to the next-ranked provider.  |
| SCHEDULED             | CANCELLED_CUSTOMER | Free until four hours before the slot; inside four hours a cancellation fee applies and the slot is released.                    |
| SCHEDULED             | CANCELLED_PROVIDER | Counts against the provider's reliability score. Three within 30 days trigger administrative review.                             |
| EN_ROUTE              | NO_SHOW            | Raised by either party no earlier than 30 minutes past the slot. A provider no-show refunds in full and penalises the provider.  |
| IN_PROGRESS           | QUOTE_REVISION     | Extra work discovered on site. The provider raises a revised quote which the customer must approve in-app before work continues. |
| AWAITING_VERIFICATION | REWORK             | Returns to IN_PROGRESS with funds still held.                                                                                    |

| From                  | To            | Rule                                                                                                      |
|-----------------------|---------------|-----------------------------------------------------------------------------------------------------------|
| AWAITING_VERIFICATION | DISPUTED      | Funds frozen pending administrative resolution; ends in PAYMENT_RELEASED, PARTIALLY_REFUNDED or REFUNDED. |
| AWAITING_VERIFICATION | AUTO_RELEASED | Seventy-two hours elapsed with no reachable customer. Rating suppressed.                                  |

## 6. Functional Requirements

Requirements are grouped by module. The Source column identifies whether the requirement originates in FR.pdf or was added by this revision; Appendix A provides the reverse mapping from the original document.

### 6.1 M1 — Service Catalogue and Pricing

The catalogue is the foundation of the system. Until services exist as records with prices and durations, nothing can be searched, booked, invoiced or reported on.

| Req. ID   | Requirement                                                                                                                                         | Priority | Source |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-CAT-01 | The administrator shall maintain service categories and the services within each category.                                                          | High     | New    |
| FR-CAT-02 | Each service shall carry a description, a base price or price band, an estimated duration, and a pricing model of flat, hourly or inspection-first. | High     | New    |
| FR-CAT-03 | A provider's expertise selections shall be bound to catalogue services; a provider shall only be offered jobs for services on which he is approved. | High     | New    |
| FR-CAT-04 | A provider may set his own price for each service he offers, within the band defined by the administrator.                                          | Medium   | New    |
| FR-CAT-05 | The administrator shall set a commission percentage globally, per category, or per provider.                                                        | High     | New    |
| FR-CAT-06 | A service may be flagged emergency-eligible, permitting same-day booking with a configured surcharge.                                               | Medium   | New    |
| FR-CAT-07 | A service may be flagged plan-eligible, warranty-eligible, or high-risk for verification routing.                                                   | Medium   | New    |

### 6.2 M2 — Customer

| Req. ID  | Requirement                                                                                                                          | Priority | Source |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-CU-01 | Customers shall register and log in to the system.                                                                                   | High     | FR.pdf |
| FR-CU-02 | Customers shall view and update their profile and change their password.                                                             | High     | FR.pdf |
| FR-CU-03 | The customer's mobile number shall be verified by OTP at registration, as it is the number to which the verification call is placed. | High     | New    |
| FR-CU-04 | The system shall provide password recovery by email or SMS one-time code.                                                            | High     | New    |
| FR-CU-05 | Customers shall save multiple addresses with area and map location, and select one per booking.                                      | High     | New    |
| FR-CU-06 | Customers shall view their booking history with status, invoice, provider and the feedback given on the verification call.           | High     | New    |

| Req. ID  | Requirement                                                                                               | Priority | Source |
|----------|-----------------------------------------------------------------------------------------------------------|----------|--------|
| FR-CU-07 | Customers may re-book a previously used provider directly from history.                                   | Low      | New    |
| FR-CU-08 | Customers may maintain a list of favourite providers.                                                     | Low      | New    |
| FR-CU-09 | Customers may deactivate their account; personal data is anonymised while financial records are retained. | Medium   | New    |

### 6.3 M3 — Service Provider

| Req. ID  | Requirement                                                                                                                                                     | Priority | Source |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-SP-01 | Service providers shall register and log in; the account remains pending until approved by the administrator.                                                   | High     | FR.pdf |
| FR-SP-02 | Service providers shall create a profile comprising name, work experience, expertise checklist, qualification, profile picture and location.                    | High     | FR.pdf |
| FR-SP-03 | Service providers shall view and update their profile and change their password.                                                                                | High     | FR.pdf |
| FR-SP-04 | Service providers shall view the ratings given by customers.                                                                                                    | High     | FR.pdf |
| FR-SP-05 | Service providers shall view the remarks given by customers.                                                                                                    | High     | FR.pdf |
| FR-SP-06 | Service providers shall upload CNIC, trade certificate and optionally a character certificate; approval shall be blocked until identity documents are verified. | High     | New    |
| FR-SP-07 | Service providers shall define a weekly availability calendar and mark days of leave; only free slots shall be offered for booking.                             | High     | New    |
| FR-SP-08 | Service providers shall define a service area as a city, a list of areas, and a working radius from a base location.                                            | High     | New    |
| FR-SP-09 | Service providers shall accept or decline an incoming job within a countdown; no response forfeits the job.                                                     | High     | New    |
| FR-SP-10 | Service providers shall view an earnings dashboard showing held, releasable, and paid amounts, commission deducted, and weekly and monthly totals.              | High     | New    |
| FR-SP-11 | Service providers shall request payout of their released wallet balance to a registered bank or mobile-wallet account.                                          | High     | New    |
| FR-SP-12 | Service providers may post one reply to a published remark; replies are visible and cannot be edited after posting.                                             | Medium   | New    |
| FR-SP-13 | Service providers shall earn tier badges computed exclusively from verified jobs.                                                                               | Low      | New    |
| FR-SP-14 | Service providers shall view their own conduct record, current demerit points and the date each point expires.                                                  | Medium   | New    |

### 6.4 M4 — Search and Matching

The filters specified in the original brief are retained; what is added is the location model that makes them operable and a ranking rule so that results are ordered rather than arbitrary.

| Req. ID  | Requirement                                                                                                                                                                                           | Priority | Source |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-SR-01 | Customers shall filter service providers by location, work experience, expertise and rating.                                                                                                          | High     | FR.pdf |
| FR-SR-02 | Customers shall view the full details of any available provider — experience, expertise, rating, customer remarks and location — in a structured layout.                                              | High     | FR.pdf |
| FR-SR-03 | Search shall begin from a service rather than a person: the customer selects the service and is shown providers offering it in the customer's area.                                                   | High     | New    |
| FR-SR-04 | Location shall be structured as city, area and map coordinates, and the distance from the customer's address shall be shown on every result.                                                          | High     | New    |
| FR-SR-05 | The system shall additionally support filtering by price range, availability today, verified-documents-only and minimum completed jobs.                                                               | Medium   | New    |
| FR-SR-06 | Default ranking shall combine rating, distance, completion rate, response speed and recent activity, with administrator-configurable weightings. Blocked and unapproved providers shall never appear. | High     | New    |
| FR-SR-07 | The system shall offer an auto-assign option in which the job is offered down the ranked list until a provider accepts.                                                                               | Medium   | New    |

## 6.5 M5 — Booking and Scheduling

| Req. ID  | Requirement                                                                                                                                                                       | Priority | Source |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-BK-01 | Customers shall book a provider by selecting a preferred date, time and required service type.                                                                                    | High     | FR.pdf |
| FR-BK-02 | Only slots free on the provider's calendar shall be selectable, and confirming a slot shall lock it against double booking.                                                       | High     | New    |
| FR-BK-03 | Customers shall describe the problem in text and may attach up to five photographs so the provider arrives prepared.                                                              | Medium   | New    |
| FR-BK-04 | The booking summary shall present the estimated price, any visit or inspection fee, and the cancellation policy before confirmation.                                              | High     | New    |
| FR-BK-05 | Customers may reschedule once without charge up to four hours before the slot.                                                                                                    | Medium   | New    |
| FR-BK-06 | Customers may cancel a booking, with the fee rules of Section 5.3 applied automatically.                                                                                          | High     | New    |
| FR-BK-07 | The system shall provide in-app messaging between customer and provider for the life of the booking, with number masking so neither party sees the other's real telephone number. | Medium   | New    |

| Req. ID  | Requirement                                                                                         | Priority | Source |
|----------|-----------------------------------------------------------------------------------------------------|----------|--------|
| FR-BK-08 | Every state change shall be written to the booking status history with actor, timestamp and reason. | High     | New    |

## 6.6 M6 — Work Execution and Evidence

The verification call is only as reliable as the record it verifies. This module produces that record.

| Req. ID  | Requirement                                                                                                                                                             | Priority | Source |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-EX-01 | The provider shall mark himself en route, notifying the customer.                                                                                                       | Medium   | New    |
| FR-EX-02 | Work shall begin only when the provider enters the start OTP displayed on the customer's phone.                                                                         | High     | New    |
| FR-EX-03 | The provider shall capture before-photographs at start and after-photographs at completion, stored against the booking with timestamps.                                 | High     | New    |
| FR-EX-04 | Parts and materials shall be logged with quantity and cost and shall appear as separate invoice lines.                                                                  | Medium   | New    |
| FR-EX-05 | Extra work shall require a revised quote approved by the customer in-app before it proceeds.                                                                            | High     | New    |
| FR-EX-06 | Completion shall generate an itemised invoice covering service, extras, parts, surcharge, discount and total.                                                           | High     | New    |
| FR-EX-07 | Each category shall carry a configurable workmanship warranty; a warranty claim shall reopen the original booking as rework rather than create a new job.               | Medium   | New    |
| FR-EX-08 | The provider shall complete a service-specific checklist, with photographic evidence where the checklist requires it, before completion can be submitted.               | High     | New    |
| FR-EX-09 | The system shall record geofenced check-in and check-out times and compare total time on site against the service's expected duration, flagging significant shortfalls. | Medium   | New    |

## 6.7 M7 — Verification and Feedback

Section 4 describes the mechanism; the following are its testable statements.

| Req. ID  | Requirement                                                                                                                     | Priority | Source |
|----------|---------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-VC-01 | Completion shall place the booking in the verification queue and freeze all associated funds.                                   | High     | New    |
| FR-VC-02 | The queue shall be ordered by waiting time and display an SLA countdown, with breaches visible on the administrative dashboard. | High     | New    |
| FR-VC-03 | The agent console shall present booking, invoice, photographs and provider history on a single screen.                          | High     | New    |

| Req. ID  | Requirement                                                                                                                                                                 | Priority | Source |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-VC-04 | Feedback shall be captured on the fixed questionnaire defined in Section 4.5 and not as free notes.                                                                         | High     | New    |
| FR-VC-05 | The agent shall record one of four outcomes: verified, verified with issue, rework required, or disputed.                                                                   | High     | New    |
| FR-VC-06 | Call attempts shall be logged individually with time, agent, duration and result; three failed attempts across two time bands shall trigger the confirmation-link fallback. | High     | New    |
| FR-VC-07 | No response within 72 hours of completion shall auto-release payment and suppress the rating.                                                                               | High     | New    |
| FR-VC-08 | A submitted verification record shall be immutable; an administrative amendment shall be stored as a linked record carrying a reason.                                       | High     | New    |
| FR-VC-09 | Calls shall be recorded following a spoken consent line, retained for a configurable period, and accessible only to administrator and finance roles.                        | Medium   | New    |
| FR-VC-10 | An agent shall not verify a booking involving a provider or customer linked to his own account.                                                                             | High     | New    |
| FR-VC-11 | The system shall route each completed job to Tier A or Tier B by the rules in Section 4.3 and store the tier applied on the booking.                                        | High     | New    |
| FR-VC-12 | A Tier B job receiving no response within 24 hours shall escalate automatically to Tier A.                                                                                  | High     | New    |

## 6.8 M8 — Payments, Escrow and Payouts

The original brief leaves money entirely outside the system. Without this module there is no commission, no transaction record, and the monthly report required by FR-RP-01 cannot be produced.

| Req. ID  | Requirement                                                                                                                                                    | Priority | Source |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-PY-01 | Customers may pay in cash on the spot or by transfer to the provider's account.                                                                                | High     | FR.pdf |
| FR-PY-02 | The system shall support online payment through a gateway, captured into platform escrow.                                                                      | High     | New    |
| FR-PY-03 | Escrow shall be released only by a passing verification outcome or the auto-release rule.                                                                      | High     | New    |
| FR-PY-04 | Cash jobs shall follow the reversed sequence of Section 4.8, with commission debited to the provider's wallet.                                                 | High     | New    |
| FR-PY-05 | A provider whose commission debt exceeds the configured ceiling shall receive no further job offers until it is cleared.                                       | High     | New    |
| FR-PY-06 | A double-entry ledger shall record every movement — capture, hold, release, commission, refund, penalty and payout. Balances shall be derived from the ledger. | High     | New    |

| Req. ID  | Requirement                                                                                                                         | Priority | Source |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-PY-07 | Refunds, whether full or partial, shall be issued to the original payment method with a linked reason.                              | High     | New    |
| FR-PY-08 | Payouts shall run on a fixed cycle producing a downloadable batch file and per-provider statements.                                 | High     | New    |
| FR-PY-09 | All payment operations shall be idempotent; a repeated gateway callback shall never release or refund twice.                        | High     | New    |
| FR-PY-10 | Coupons and referral credits shall apply at booking and shall be settled against platform commission rather than provider earnings. | Low      | New    |

## 6.9 M9 — Ratings and Reputation

| Req. ID  | Requirement                                                                                                                                              | Priority | Source |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-RT-01 | Customers shall rate the provider after completion according to their satisfaction.                                                                      | High     | FR.pdf |
| FR-RT-02 | Customers shall give remarks and comments about the provider after completion.                                                                           | High     | FR.pdf |
| FR-RT-03 | Ratings and remarks shall originate only from a completed verification call or its confirmation-link fallback; no direct in-app rating form shall exist. | High     | New    |
| FR-RT-04 | The published score shall be the average of the four criteria, weighted toward the most recent twenty jobs.                                              | Medium   | New    |
| FR-RT-05 | Profiles shall display the score, the number of jobs it is based on, and the distribution of ratings.                                                    | Medium   | New    |
| FR-RT-06 | Remarks shall be published under the customer's first name and initial; full identity shall never be shown to the provider.                              | Medium   | New    |
| FR-RT-07 | The administrator may unpublish an abusive remark; the underlying rating remains in the score and the removal is logged.                                 | Medium   | New    |

## 6.10 M10 — Complaints and Disputes

| Req. ID  | Requirement                                                                                                                        | Priority | Source |
|----------|------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-CP-01 | Customers shall register a complaint to the administrator about a provider in the event of a serious issue such as misbehaviour.   | High     | FR.pdf |
| FR-CP-02 | The administrator shall view customer complaints and take appropriate action.                                                      | High     | FR.pdf |
| FR-CP-03 | A complaint shall have the states open, under review, awaiting response, resolved and rejected, with an SLA by severity.           | High     | New    |
| FR-CP-04 | Complaints shall attach to a booking, carry photographic evidence, and be visible to the verification agent handling that booking. | Medium   | New    |

| Req. ID  | Requirement                                                                                                                              | Priority | Source |
|----------|------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-CP-05 | The provider shall be given a right of reply before any penalty is applied.                                                              | High     | New    |
| FR-CP-06 | Resolution outcomes shall be no action, warning, partial refund, full refund, provider penalty, temporary suspension or permanent block. | High     | New    |
| FR-CP-07 | Providers shall also be able to raise complaints — unsafe premises, non-payment, or abusive conduct by a customer.                       | Medium   | New    |
| FR-CP-08 | A complaint marked as a safety issue shall escalate immediately to the top of the administrative queue.                                  | High     | New    |

### 6.11 M11 — Notifications

Absent from the original brief. Templates are administrator-editable and stored per event and per channel.

| Req. ID  | Requirement                                                                                                                                      | Priority | Source |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-NT-01 | The system shall notify by email, SMS and in-app message on every booking state change.                                                          | High     | New    |
| FR-NT-02 | Providers shall be alerted on new job offer with countdown, acceptance confirmation, upcoming job reminder and payment release.                  | High     | New    |
| FR-NT-03 | Customers shall be alerted on booking confirmation, provider en route, start OTP, work completed, pending verification call and payment release. | High     | New    |
| FR-NT-04 | Administrators shall be alerted on new provider awaiting approval, verification SLA breach, dispute opened and safety-flagged complaint.         | Medium   | New    |
| FR-NT-05 | Every notification shall be logged with channel, recipient, template and delivery status.                                                        | Medium   | New    |

### 6.12 M12 — Administration and Operations

| Req. ID  | Requirement                                                                      | Priority | Source |
|----------|----------------------------------------------------------------------------------|----------|--------|
| FR-AD-01 | The administrator shall log in to the system.                                    | High     | FR.pdf |
| FR-AD-02 | The administrator shall approve registered service providers.                    | High     | FR.pdf |
| FR-AD-03 | The administrator shall manage service providers — add, view, update and delete. | High     | FR.pdf |
| FR-AD-04 | The administrator shall update provider details and manage login passwords.      | High     | FR.pdf |
| FR-AD-05 | The administrator shall manage customers — add, view, update and delete.         | High     | FR.pdf |

| Req. ID  | Requirement                                                                                                                                                                                 | Priority | Source |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-AD-06 | The administrator shall view the ratings given to each provider and each provider's average rating.                                                                                         | High     | FR.pdf |
| FR-AD-07 | The administrator shall view the remarks given by customers about each provider.                                                                                                            | High     | FR.pdf |
| FR-AD-08 | The administrator shall block a provider in the event of continuous poor rating.                                                                                                            | High     | FR.pdf |
| FR-AD-09 | Deletion shall be implemented as soft deactivation: the account cannot log in or be booked, while all bookings, ratings and ledger entries survive. Physical deletion shall not be offered. | High     | New    |
| FR-AD-10 | Approval shall require a document check; the reviewer, decision and reason shall be recorded, and rejection shall notify the provider with an explanation.                                  | High     | New    |
| FR-AD-11 | An automatic rule shall flag providers whose rolling average falls below a configured threshold.                                                                                            | Medium   | New    |
| FR-AD-12 | An operations board shall display today's bookings by state, verification queue depth, SLA breaches, open disputes and total escrow held.                                                   | Medium   | New    |
| FR-AD-13 | The system shall support role and permission management for agent, finance and administrator logins.                                                                                        | High     | New    |
| FR-AD-14 | An append-only audit log shall record every privileged action, filterable by actor, entity and date.                                                                                        | High     | New    |
| FR-AD-15 | System settings shall include commission rates, cancellation fees, SLA timers, calling hours, auto-release window, surcharges, verification tier thresholds and penalty values.             | High     | New    |

### 6.13 M13 — Maintenance Plans

Recurring maintenance is the natural extension of a system named for home maintenance, and converts one-off jobs into predictable revenue.

| Req. ID  | Requirement                                                                                                                       | Priority | Source |
|----------|-----------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-MP-01 | The administrator shall define maintenance plans, such as a quarterly plumbing inspection with two included call-outs.            | Medium   | New    |
| FR-MP-02 | A subscribed customer's visits shall be scheduled automatically and offered first to the same provider for continuity.            | Medium   | New    |
| FR-MP-03 | Plan visits shall follow the identical verification flow, with release drawing on the plan balance rather than a per-job payment. | Medium   | New    |
| FR-MP-04 | Customers shall view remaining entitlements and renewal date, and may cancel a plan.                                              | Medium   | New    |

### 6.14 M14 — Reporting

| Req. ID  | Requirement                                                                                                                                              | Priority | Source |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-RP-01 | The administrator shall generate a monthly report of total bookings and their successful handling.                                                       | High     | FR.pdf |
| FR-RP-02 | Successful handling shall be defined as verified or auto-released without dispute, so that the figure is computable from system data.                    | High     | New    |
| FR-RP-03 | A revenue report shall present gross booking value, commission earned, refunds, penalties and net, by month and by category.                             | High     | New    |
| FR-RP-04 | A provider performance report shall present jobs, acceptance rate, cancellation rate, average rating, rework rate and disputes.                          | Medium   | New    |
| FR-RP-05 | A verification report shall present calls placed, first-attempt success rate, average call duration, outcome mix, SLA compliance and auto-release count. | Medium   | New    |
| FR-RP-06 | Every report shall export to PDF and Excel and shall be printable.                                                                                       | Medium   | New    |

## 6.15 M15 — Conduct and Penalties

Section 7 defines the framework; the following are its system requirements.

| Req. ID  | Requirement                                                                                                                                                      | Priority | Source |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FR-PN-01 | The system shall record demerit points against a provider for verified breaches, in accordance with the schedule in Section 7.2.                                 | High     | New    |
| FR-PN-02 | Demerit points shall expire on a rolling 180-day window and shall decay by one point for every 30 days without a new breach.                                     | High     | New    |
| FR-PN-03 | Accumulated points shall trigger the graduated consequences defined in Section 7.3 automatically.                                                                | High     | New    |
| FR-PN-04 | Financial penalties shall be debited from the provider's wallet; where the balance is insufficient the shortfall shall become a debt that blocks new job offers. | High     | New    |
| FR-PN-05 | Total financial liability for a single job shall be capped at the job value plus a configured maximum fine.                                                      | High     | New    |
| FR-PN-06 | No penalty shall be applied until the provider has been shown the evidence and given 48 hours to respond.                                                        | High     | New    |
| FR-PN-07 | A provider may appeal any penalty to the administrator; the appeal and its decision shall be recorded in the audit log.                                          | Medium   | New    |
| FR-PN-08 | The penalty schedule shall be presented to the provider at registration and shall be visible in his conduct record thereafter.                                   | High     | New    |
| FR-PN-09 | Providers in the top performance tier shall receive priority job offers, a reduced commission rate and accelerated payouts.                                      | Low      | New    |

## 7. Conduct and Penalty Framework

### 7.1 Principles

5. Penalties follow verified evidence, never a single unsupported complaint. One dissatisfied customer is not proof of misconduct.
6. Integrity breaches are separated from quality shortfalls. A slow or careless tradesman is coached through a graduated ladder; a tradesman who demanded cash beyond the approved quote is removed.
7. No penalty is applied without a right of reply. The provider is shown the evidence and given 48 hours to respond before any money moves.
8. Penalties are bounded. Unlimited liability drives competent tradesmen off the platform faster than any competitor.
9. The framework is published. A penalty the provider did not know existed reads as arbitrary, and word travels quickly among tradesmen.

### 7.2 Demerit Schedule

| Breach                                                     | Points | Financial consequence                                |
|------------------------------------------------------------|--------|------------------------------------------------------|
| Late arrival exceeding 30 minutes without notice           | 1      | —                                                    |
| Cancellation inside the four-hour window                   | 3      | Cancellation fee debited                             |
| Rework required, verified                                  | 4      | Rework carried out at own cost                       |
| Poor rating streak — three consecutive jobs below 3.0      | 5      | Ranking demotion                                     |
| No-show                                                    | 8      | Full customer refund plus fixed penalty              |
| Second failed rework on the same job                       | 8      | Partial or full refund                               |
| Sending an unregistered substitute worker                  | 20     | Suspension                                           |
| Taking the customer off-platform                           | 20     | Commission recovery plus suspension                  |
| Charging beyond the approved quote                         | 25     | Refund of the excess plus a fine of twice the excess |
| Falsified completion — marked complete, work not performed | 25     | Full refund plus suspension                          |
| Unsafe work causing damage to property                     | 25     | Damage recovery plus suspension                      |
| Misbehaviour or harassment, verified                       | 30     | Immediate permanent block                            |
| Fabricated ratings or colluding accounts                   | 30     | Permanent block, wallet balance forfeited            |

### 7.3 Thresholds

| Accumulated points | Consequence                                                |
|--------------------|------------------------------------------------------------|
| 10                 | Written warning, visible in the provider's conduct record. |

| Accumulated points                                                            | Consequence                                                                  |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 20                                                                            | Search-ranking demotion for 30 days — fewer job offers, felt immediately.    |
| 30                                                                            | Seven-day suspension; no job offers issued.                                  |
| 45                                                                            | Thirty-day suspension and re-verification of documents before reinstatement. |
| 60, or any single breach carrying 25 points or more in the integrity category | Permanent block.                                                             |

## 7.4 Financial Mechanics

- Penalties are debited from the provider's wallet and are never invoiced separately. Where the balance is insufficient, the shortfall becomes a debt that suspends new job offers until cleared — an enforcement mechanism that costs the platform nothing to operate.
- Total liability arising from a single job is capped at the job value plus a configured maximum fine.
- A refundable security deposit may be required at onboarding for high-value categories, returned on exit where the conduct record is clean.

## 7.5 Counterweights

A purely punitive framework loses supply, and in this market supply is the scarce side. The following balance it:

- Points decay with clean work, and the provider can see his conduct record and his progress back toward zero.
- Top-tier providers receive priority job offers, a reduced commission rate and accelerated payouts.
- Badges earned from verified jobs feed back into search ranking.
- Every penalty is appealable to the administrator, and the decision is recorded.

## 8. Use Cases

### 8.1 Actors

| Actor                 | Role in the system                                                                    |
|-----------------------|---------------------------------------------------------------------------------------|
| Customer              | Books services, receives the verification call, raises complaints.                    |
| Service provider      | Publishes skills and availability, performs work, submits evidence, receives payment. |
| Verification agent    | Places verification calls and records structured feedback.                            |
| Finance officer       | Releases escrow, processes refunds, runs payouts.                                     |
| Administrator         | Approves and blocks providers, resolves disputes, configures and reports.             |
| Payment gateway       | External actor: captures, holds, releases and refunds funds.                          |
| SMS and email gateway | External actor: delivers one-time codes and notifications.                            |

### 8.2 Use Case Index

| ID    | Use case                                            | Primary actor                     |
|-------|-----------------------------------------------------|-----------------------------------|
| UC-01 | Register and verify a customer account              | Customer                          |
| UC-02 | Register as a service provider and submit documents | Service provider                  |
| UC-03 | Approve or reject a provider application            | Administrator                     |
| UC-04 | Search for a service and view providers             | Customer                          |
| UC-05 | Book a service provider for a slot                  | Customer                          |
| UC-06 | Accept or decline a job offer                       | Service provider                  |
| UC-07 | Start work using the customer's OTP                 | Service provider                  |
| UC-08 | Raise a revised quote for extra work                | Service provider                  |
| UC-09 | Submit completion with evidence                     | Service provider                  |
| UC-10 | Verify completed work by telephone                  | Verification agent                |
| UC-11 | Release payment and publish rating                  | Finance officer                   |
| UC-12 | Handle an unreachable customer                      | System                            |
| UC-13 | Raise and resolve a complaint                       | Customer, Administrator           |
| UC-14 | Resolve a payment dispute                           | Administrator                     |
| UC-15 | Apply a penalty and hear an appeal                  | Administrator                     |
| UC-16 | Request and process a payout                        | Service provider, Finance officer |
| UC-17 | Generate the monthly report                         | Administrator                     |

### 8.3 Detailed Use Cases

#### UC-05 — Book a service provider for a slot

| Element         | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor           | Customer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Precondition    | The customer is logged in, has a verified mobile number and at least one saved address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Trigger         | The customer selects a service from the catalogue.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Main flow       | 1. The customer selects the service and the address. 2. The system lists providers offering that service in the area, ranked per FR-SR-06. 3. The customer opens a provider's profile and reviews experience, rating, remarks and distance. 4. The customer selects an available slot from the provider's calendar. 5. The customer describes the problem and may attach photographs. 6. The system presents the estimated price, any visit fee and the cancellation policy. 7. The customer confirms; the slot is locked and the booking enters REQUESTED. 8. The provider is notified with an acceptance countdown. |
| Alternate flows | 4a. No slot is available — the customer selects a different provider or a later date. 7a. The customer chooses auto-assign — the system offers the job down the ranked list until a provider accepts.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Exception flows | 7b. The slot is taken concurrently — the system reports the clash and refreshes availability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Postcondition   | A booking exists in REQUESTED and the slot is held.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Requirements    | FR-BK-01 to FR-BK-04, FR-SR-03, FR-SR-06, FR-SR-07                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### UC-10 — Verify completed work by telephone

| Element         | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor           | Verification agent                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Precondition    | A booking is in AWAITING_VERIFICATION, routed to Tier A, and all funds are frozen.                                                                                                                                                                                                                                                                                                                                                                                           |
| Trigger         | The agent takes the next booking from the verification queue.                                                                                                                                                                                                                                                                                                                                                                                                                |
| Main flow       | 1. The console presents the booking, invoice, before and after photographs, and the provider's history. 2. The agent places the call and reads the recording-consent line. 3. The agent confirms the customer's identity against the booking. 4. The agent works through the fixed questionnaire of Section 4.5. 5. The agent selects one of the four outcomes. 6. The system writes an immutable verification record and moves the booking to VERIFIED, REWORK or DISPUTED. |
| Alternate flows | 2a. No answer — the attempt is logged and the booking returns to the queue for a later time band. 5a. Verified with issue — a complaint ticket opens automatically and the provider is flagged.                                                                                                                                                                                                                                                                              |
| Exception flows | 3a. The person answering is not the customer — the agent terminates and reschedules the attempt. 1a. The agent is linked to the provider or customer — the console refuses the booking under FR-VC-10.                                                                                                                                                                                                                                                                       |
| Postcondition   | A verification record exists; the booking has moved to an outcome state; funds remain frozen unless the outcome permits release.                                                                                                                                                                                                                                                                                                                                             |
| Requirements    | FR-VC-01 to FR-VC-06, FR-VC-08 to FR-VC-11                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**UC-12 — Handle an unreachable customer**

| Element         | Detail                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor           | System                                                                                                                                                                                                                                                                                                                                                                                                   |
| Precondition    | A booking has been in AWAITING_VERIFICATION with failed contact attempts.                                                                                                                                                                                                                                                                                                                                |
| Trigger         | A scheduled job evaluates the verification queue.                                                                                                                                                                                                                                                                                                                                                        |
| Main flow       | 1. After three failed attempts across two time bands within 24 hours, the system sends an SMS and WhatsApp confirmation link with a one-time code. 2. If the customer confirms through the link, the outcome is recorded as verified and payment releases. 3. If there is no response by 72 hours after completion, the system auto-releases payment, records the completion, and suppresses the rating. |
| Alternate flows | 3a. The customer responds within the seven-day post-release window with a complaint — a complaint ticket opens and follows UC-13.                                                                                                                                                                                                                                                                        |
| Postcondition   | The provider has been paid; no rating has been published; the booking is marked AUTO_RELEASED.                                                                                                                                                                                                                                                                                                           |
| Requirements    | FR-VC-06, FR-VC-07, FR-PY-03                                                                                                                                                                                                                                                                                                                                                                             |

**UC-14 — Resolve a payment dispute**

| Element         | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor           | Administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Precondition    | A booking is in DISPUTED and funds are frozen.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Trigger         | The administrator opens the dispute queue.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Main flow       | 1. The administrator reviews the evidence floor — OTP, geofence times, photographs, checklist, invoice — alongside the verification record. 2. The provider is shown the evidence and given 48 hours to reply. 3. The administrator selects a resolution: full release, partial release, full refund, or refund with a penalty. 4. The ledger is updated and both parties are notified. 5. Where a penalty applies, demerit points are recorded per Section 7.2. |
| Alternate flows | 2a. The provider does not reply within 48 hours — the administrator proceeds on the evidence available.                                                                                                                                                                                                                                                                                                                                                          |
| Exception flows | 3a. The provider's wallet is insufficient for the penalty — the shortfall becomes a debt blocking new job offers under FR-PN-04.                                                                                                                                                                                                                                                                                                                                 |
| Postcondition   | The dispute is closed, funds have moved, and the audit log records the decision.                                                                                                                                                                                                                                                                                                                                                                                 |
| Requirements    | FR-CP-05, FR-CP-06, FR-PY-06, FR-PY-07, FR-PN-01, FR-PN-04, FR-PN-06                                                                                                                                                                                                                                                                                                                                                                                             |

## 9. External Interface Requirements

### 9.1 User Interfaces

The following screens constitute the minimum interface. All are mobile-first and must remain usable at 360 px width.

| Area               | Screens                                                                                                                                                                                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Public             | Home and service categories; service listing; provider profile; registration; login; password recovery.                                                                                                                                                                                                   |
| Customer           | Dashboard; search and filter results; booking form with slot picker; booking summary and confirmation; booking detail with live status; booking history; invoice view; address book; profile and password; complaint form; maintenance plans.                                                             |
| Service provider   | Dashboard with today's jobs; job offer with countdown; job detail; start-work OTP entry; checklist and photograph capture; revised quote form; completion submission; earnings and wallet; payout request; availability calendar; service and price list; documents; ratings and remarks; conduct record. |
| Verification agent | Verification queue with SLA countdown; agent console combining booking, invoice, photographs and provider history with the questionnaire; call attempt log.                                                                                                                                               |
| Finance            | Escrow held; release queue; refunds; payout batch; cash reconciliation; commission debt list.                                                                                                                                                                                                             |
| Administrator      | Operations board; provider approval queue; provider and customer management; catalogue management; complaint queue; dispute queue; penalty and appeal management; roles and permissions; settings; audit log; reports.                                                                                    |

### 9.2 Hardware Interfaces

- Camera access on the provider's device for before and after photographs and checklist evidence.
- Location services on the provider's device for geofenced check-in and check-out.
- Telephone handset or softphone at the agent workstation, with call recording.

### 9.3 Software Interfaces

| Interface                     | Purpose                                                                   | Requirement                                                                         |
|-------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Payment gateway               | Capture, hold, release and refund funds; supports card and mobile wallet. | Must support delayed capture or held funds, and must deliver idempotent callbacks.  |
| SMS gateway                   | One-time codes, booking notifications, verification-link fallback.        | Delivery status must be returned and logged.                                        |
| Email service                 | Notifications, invoices, reports.                                         | Templated, with delivery logging.                                                   |
| Mapping and geocoding service | Address coordinates, distance calculation, geofence evaluation.           | Must return coordinates for a text address and compute distance between two points. |
| Messaging platform (optional) | WhatsApp confirmation link for Tier B verification.                       | Optional; SMS is the mandatory fallback.                                            |

## 9.4 Communication Interfaces

- All client-server traffic uses HTTPS.
- Gateway callbacks are authenticated and verified against a signature before processing.
- Notification dispatch and report generation run on an asynchronous queue rather than inside the web request.

## 10. Data Requirements

### 10.1 Entity Groups

| Group             | Entities                                                                                                        |
|-------------------|-----------------------------------------------------------------------------------------------------------------|
| Identity          | users, roles, user_roles, customers, providers, provider_documents, otp_tokens, password_resets                 |
| Catalogue         | categories, services, provider_services, commission_rules                                                       |
| Place and time    | cities, areas, addresses, provider_service_areas, provider_availability, provider_time_off                      |
| Jobs              | bookings, booking_items, booking_status_history, job_evidence, job_checklist_results, quote_revisions, invoices |
| Verification      | verification_calls, verification_call_attempts, verification_responses                                          |
| Money             | payments, ledger_entries, wallets, payouts, refunds, coupons                                                    |
| Trust and conduct | ratings, remarks, complaints, complaint_events, disputes, penalties, demerit_points                             |
| Platform          | subscriptions, plan_entitlements, notifications, notification_templates, settings, audit_log                    |

### 10.2 Selected Data Dictionary

The following entities carry the logic that distinguishes this system and are specified in detail. Remaining entities follow conventional structure.

#### bookings

| Field                                | Type     | Notes                                                    |
|--------------------------------------|----------|----------------------------------------------------------|
| booking_id                           | PK       | —                                                        |
| customer_id, provider_id, service_id | FK       | provider_id is null while a booking is in auto-assign.   |
| address_id                           | FK       | The customer's saved address for this job.               |
| scheduled_start, scheduled_end       | datetime | The locked slot.                                         |
| status                               | enum     | One of the states in Section 5.2.                        |
| is_emergency                         | boolean  | Drives the surcharge.                                    |
| quoted_amount, final_amount          | decimal  | Final amount is set at completion.                       |
| payment_mode                         | enum     | cash or online — determines the sequence in Section 4.8. |
| verification_tier                    | enum     | A or B, assigned at completion per FR-VC-11.             |
| start_otp_verified_at                | datetime | Null until the provider enters the start OTP.            |
| checkin_at, checkout_at              | datetime | Geofenced times.                                         |
| created_at, updated_at               | datetime | —                                                        |

**verification\_calls**

| Field                                      | Type         | Notes                                                           |
|--------------------------------------------|--------------|-----------------------------------------------------------------|
| verification_call_id                       | PK           | —                                                               |
| booking_id                                 | FK, unique   | One verification record per booking.                            |
| tier                                       | enum         | A or B.                                                         |
| agent_id                                   | FK           | Null for Tier B automated confirmations.                        |
| outcome                                    | enum         | verified, verified_with_issue, rework, disputed, auto_released. |
| consent_to_release                         | boolean      | Recorded verbally on the call.                                  |
| extra_charge_demanded                      | boolean      | The integrity question of Section 4.5.                          |
| quality, punctuality, conduct, cleanliness | int 1–5      | Averaged to produce the published rating.                       |
| remark_text                                | text         | Transcribed by the agent.                                       |
| recording_ref, call_duration_seconds       | varchar, int | —                                                               |
| submitted_at                               | datetime     | The record becomes immutable at this moment.                    |

**ledger\_entries**

| Field            | Type     | Notes                                                                     |
|------------------|----------|---------------------------------------------------------------------------|
| entry_id         | PK       | —                                                                         |
| booking_id       | FK       | Null for payout and adjustment entries.                                   |
| account_type     | enum     | customer, provider_wallet, platform_escrow, platform_commission, penalty. |
| account_id       | FK       | The party the account belongs to.                                         |
| direction        | enum     | debit or credit.                                                          |
| amount           | decimal  | Positive value; direction carries the sign.                               |
| reason           | enum     | capture, hold, release, commission, refund, penalty, payout, adjustment.  |
| related_entry_id | FK       | Links the two halves of a double-entry pair.                              |
| created_at       | datetime | Entries are never updated or deleted.                                     |

**10.3 Integrity Rules**

10. A ratings row may exist only where a verification\_call\_id is present and that call carries a completed outcome. This is enforced by a foreign key and a not-null constraint rather than by application code; it is the single database rule that makes every rating on the platform trustworthy.
11. A booking may not move to WORK\_COMPLETED unless start\_otp\_verified\_at is set.
12. A booking may not move to PAYMENT\_RELEASED unless a verification record exists with an outcome permitting release, or the auto-release condition of FR-VC-07 is satisfied.

13. Every ledger entry is paired; the sum of debits equals the sum of credits for any booking at rest.
14. No user row participating in a booking may be physically deleted; deactivation is a status change.
15. audit\_log and verification\_calls permit insert only. Amendments are new rows referencing the original.

## 11. Non-Functional Requirements

### 11.1 Security

| Req. ID   | Requirement                                                                                                                                   | Priority | Source |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| NFR-SE-01 | Passwords shall be stored using a modern adaptive hash such as bcrypt or Argon2, never encrypted or plain.                                    | High     | New    |
| NFR-SE-02 | Role-based authorisation shall be enforced on the server for every request; hiding interface elements shall not be treated as access control. | High     | New    |
| NFR-SE-03 | All database access shall use parameterised queries.                                                                                          | High     | New    |
| NFR-SE-04 | State-changing forms shall carry CSRF tokens, and all output shall be escaped against cross-site scripting.                                   | High     | New    |
| NFR-SE-05 | Uploads shall be restricted by file type and size and stored outside the web root.                                                            | High     | New    |
| NFR-SE-06 | Login, OTP request and OTP verification endpoints shall be rate-limited.                                                                      | High     | New    |

### 11.2 Privacy

| Req. ID   | Requirement                                                                                                                           | Priority | Source |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| NFR-PR-01 | Telephone numbers shall be masked between customer and provider; neither party shall see the other's real number.                     | High     | New    |
| NFR-PR-02 | Call recordings shall be encrypted at rest, retained for a configured period, and accessible only to administrator and finance roles. | High     | New    |
| NFR-PR-03 | Identity documents shall be visible only to the approving administrator.                                                              | High     | New    |
| NFR-PR-04 | Customer remarks shall be published under first name and initial only.                                                                | Medium   | New    |

### 11.3 Integrity and Reliability

| Req. ID   | Requirement                                                                | Priority | Source |
|-----------|----------------------------------------------------------------------------|----------|--------|
| NFR-IN-01 | All monetary operations shall execute within database transactions.        | High     | New    |
| NFR-IN-02 | Payment gateway callbacks shall be idempotent.                             | High     | New    |
| NFR-IN-03 | The audit log shall be append-only.                                        | High     | New    |
| NFR-IN-04 | The database shall be backed up daily with a documented restore procedure. | High     | New    |

### 11.4 Performance

| Req. ID   | Requirement                                                                                                         | Priority | Source |
|-----------|---------------------------------------------------------------------------------------------------------------------|----------|--------|
| NFR-PE-01 | Search results shall return within 2 seconds for a dataset of 10,000 providers.                                     | High     | New    |
| NFR-PE-02 | Indexes shall exist on city, area, service, booking status and provider rating.                                     | High     | New    |
| NFR-PE-03 | Notification dispatch and report generation shall run asynchronously on a queue.                                    | Medium   | New    |
| NFR-PE-04 | The provider-facing interface shall remain usable on a 3G connection, and image upload shall tolerate interruption. | High     | New    |

### 11.5 Usability and Accessibility

| Req. ID   | Requirement                                                                            | Priority | Source |
|-----------|----------------------------------------------------------------------------------------|----------|--------|
| NFR-US-01 | The interface shall be mobile-first and fully usable at 360 px width.                  | High     | New    |
| NFR-US-02 | The interface shall support English and Urdu, including right-to-left rendering.       | Medium   | New    |
| NFR-US-03 | Form errors shall be shown as visible text adjacent to the field, not by colour alone. | Medium   | New    |
| NFR-US-04 | Keyboard focus shall be visible on every interactive element.                          | Medium   | New    |

### 11.6 Maintainability and Operability

| Req. ID   | Requirement                                                                                                                                      | Priority | Source |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| NFR-MA-01 | All timers, fees, thresholds, commission rates and windows shall be held in configuration and editable by the administrator without code change. | High     | New    |
| NFR-MA-02 | Seed data shall be provided for categories, services, cities and areas to permit demonstration.                                                  | Medium   | New    |
| NFR-MA-03 | Application errors shall be logged with sufficient context for diagnosis without recording sensitive data.                                       | Medium   | New    |

## 12. Out of Scope and Future Enhancements

The following are recognised as current practice in this category but are deliberately excluded from this release. Recording the decision is more valuable than attempting the feature.

| Feature                           | Reason for exclusion                                                                                                                                                                                       |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI-ranked matching                | The weighted ranking formula of FR-SR-06 delivers most of the benefit. Revisit once sufficient completed-job data exists.                                                                                  |
| Live GPS tracking of the provider | Requires a native application with background location. The EN_ROUTE state with an ETA notification answers the customer's actual question.                                                                |
| Video quotation calls             | A second real-time system layered on the verification queue; disproportionate for a first release.                                                                                                         |
| Dynamic surge pricing             | A fixed emergency surcharge is explainable and defensible. Surge requires demand data not available at launch.                                                                                             |
| Insurance-backed guarantee        | A commercial arrangement rather than a software feature. The workmanship warranty of FR-EX-07 is the buildable portion.                                                                                    |
| Retention against warranty period | Holding back a percentage of payment until the warranty expires is a sound extension of the escrow model, deferred to keep the first payment implementation simple.                                        |
| IoT and sensor-based verification | Verifies the outcome rather than the visit — for example the temperature differential following AC servicing. Narrow in application but a natural extension for a system named for smart home maintenance. |
| Multi-city operations             | The schema is city-aware from the outset; operations launch in a single city.                                                                                                                              |
| Native mobile applications        | The specification targets a responsive web application. Native applications would follow once volume justifies them.                                                                                       |

## 13. Implementation Plan

Sequenced so that each phase leaves a demonstrable system. Phase 3 carries the distinguishing functionality and should not be deferred to the end of the schedule.

| Phase                       | Modules         | Deliverable                                                                                                                                                                                                                                                 |
|-----------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 — Foundation              | M1, M2, M3, M12 | Authentication with roles, customer and provider profiles, document upload, approval and blocking, the service catalogue, settings and audit log. Ends with an administrator able to onboard a real tradesman.                                              |
| 2 — Transacting             | M4, M5, M6      | Availability calendar, service-first search with ranking, slot booking, acceptance and decline, the full state machine, start OTP, checklists, evidence photographs and the itemised invoice. Ends with a job that runs end to end and stops at completion. |
| 3 — Verification and money  | M7, M8, M9      | Verification queue, agent console, questionnaire, four outcomes, attempt logging, tier routing, auto-release, escrow ledger, commission, payouts, and ratings published from verified calls only. Ends with money that moves correctly.                     |
| 4 — Trust and communication | M10, M11, M15   | Complaint and dispute lifecycles, right of reply, the penalty framework and appeals, and the notification layer across every state change.                                                                                                                  |
| 5 — Depth                   | M13, M14        | Maintenance plans, the five reports with PDF and Excel export, and the live operations board.                                                                                                                                                               |

## Appendix A — Traceability to FR.pdf

Every requirement stated in the original functional requirements document is mapped below to its identifier in this specification. No original requirement has been dropped.

| Requirement in FR.pdf                                                                                  | Module | SRS requirement    |
|--------------------------------------------------------------------------------------------------------|--------|--------------------|
| Service provider can register and login                                                                | M3     | FR-SP-01           |
| Service provider can make their profile: name, experience, expertise, qualification, picture, location | M3     | FR-SP-02           |
| Service provider can view/update profile and update password                                           | M3     | FR-SP-03           |
| Service provider can check the rating given by the customer                                            | M3     | FR-SP-04           |
| Service provider can check any remarks given by the customer                                           | M3     | FR-SP-05           |
| Customers can register and login                                                                       | M2     | FR-CU-01           |
| Customers can view/update profile and update password                                                  | M2     | FR-CU-02           |
| Searching of service providers by location, experience, expertise, rating                              | M4     | FR-SR-01           |
| View service provider details in a proper layout                                                       | M4     | FR-SR-02           |
| Book any service provider by date, time and service type                                               | M5     | FR-BK-01           |
| Rating of service provider after completion                                                            | M9     | FR-RT-01           |
| Remarks about service provider after completion                                                        | M9     | FR-RT-02           |
| Complaint about service provider to the admin                                                          | M10    | FR-CP-01           |
| Admin can login                                                                                        | M12    | FR-AD-01           |
| Admin approves registered service providers                                                            | M12    | FR-AD-02           |
| Admin can manage service providers (add/view/update/delete)                                            | M12    | FR-AD-03, FR-AD-09 |
| Admin can update provider details and manage passwords                                                 | M12    | FR-AD-04           |
| Admin can manage customers (add/view/update/delete)                                                    | M12    | FR-AD-05, FR-AD-09 |
| Admin can check ratings and average rating per provider                                                | M12    | FR-AD-06           |
| Admin can check remarks per provider                                                                   | M12    | FR-AD-07           |
| Admin can block a provider for continuous poor rating                                                  | M12    | FR-AD-08           |
| Admin can view complaints and take action                                                              | M10    | FR-CP-02           |

| Requirement in FR.pdf                                                      | Module | SRS requirement                  |
|----------------------------------------------------------------------------|--------|----------------------------------|
| Admin can generate a monthly report of bookings and successful handling    | M14    | FR-RP-01, FR-RP-02               |
| Payment on the spot or by transfer to the provider's account               | M8     | FR-PY-01                         |
| Responsibility for verifying the completed service rests with the customer | M7     | FR-VC-01 to FR-VC-12 (Section 4) |

**Note on the final row.** The original document assigns verification responsibility to the customer without defining a mechanism. Section 4 of this specification supplies that mechanism, and it is the principal enhancement of this revision.

## Appendix B — Summary of Changes from FR.pdf

| Area                        | Original                                            | This specification                                                                              |
|-----------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Modules                     | 4                                                   | 15                                                                                              |
| Roles                       | 3 — customer, service provider, admin               | 5 — the above plus verification agent and finance officer                                       |
| Functional requirements     | 24 stated                                           | 24 carried forward, 106 added — 130 in total                                                    |
| Non-functional requirements | None stated                                         | 25, grouped into six areas                                                                      |
| Service definition          | Expertise as a free checklist; no bookable services | A structured catalogue of six categories with pricing models, durations and attributes          |
| Scheduling                  | Customer selects any date and time                  | Provider availability calendar with slot locking against double booking                         |
| Booking states              | None                                                | A ten-state machine with eight branch transitions and a full status history                     |
| Work verification           | Assigned to the customer, no mechanism              | Evidence floor plus two-tier verification, with a staffed telephone call gating payment release |
| Payments                    | Cash or transfer, outside the system                | Escrow, ledger, commission, refunds, payouts, and a defined cash path                           |
| Ratings                     | Customer rates after completion                     | Ratings originate only from a verified call, enforced at database level                         |
| Complaints                  | Registered; admin takes action                      | A five-state lifecycle with SLA, evidence, right of reply and defined outcomes                  |
| Provider discipline         | Block for poor rating                               | A demerit framework with a published schedule, graduated thresholds, appeals and counterweights |
| Notifications               | None                                                | Email, SMS and in-app across every state change, with delivery logging                          |